Game Theory & Strategy

Week-by-Week High-Weightage Practice

Targeted Problems for Maximum Scoring

Based on Aug 2025 & Dec 2025 End-Term Analysis

IITM BS Exam Preparation

August 2026

Exam Weightage Analysis (from 2 past papers)

Topic	Week	Marks (approx)	Difficulty
Coalition function + Shapley-Shubik	W10	**8–10**	Easy–Medium
Network effects (all sub-parts)	W9	**6–7**	Medium
VCG ad auctions	W12	**5–6**	Easy
Cournot duopoly	W2	**4–5**	Easy
First-price auction + order stats	W4	**4–5**	Easy
Contested garment	W11	**2–3**	Easy
Bayes' theorem	W6	**4**	Easy
Cable TV cost allocation	W11	**3–6**	Medium
Market entry + mixed NE	W2	**3–4**	Medium–Hard
Declining industry / exit game	W2/W3	3–5	Medium
Roommate cleaning game	W2	3	Medium
Conceptual MCQs (matching, cascades)	W6/W7	2–3	Easy

Strategy: The shaded rows account for **35+ marks out of 45**. Master these and you're set.

Contents

1 Week 2: Strategic Form Games

Cournot Duopoly — ˜5 marks every exam

Master the three formulas and you'll never lose marks here.

Core Formulas — Cournot with Zero Costs

Demand: $p = a - Q$, where $Q = q_1 + q_2$. Zero marginal cost.
Best Response: $BR_i(q_j) = \dfrac{a - q_j}{2}$
Nash Equilibrium: $q_1^* = q_2^* = \dfrac{a}{3}$
Equilibrium Price: $p^* = \dfrac{a}{3}$
Equilibrium Profit: $\pi_i^* = \left(\dfrac{a}{3}\right)^2$
With symmetric cost c: Replace a with $(a - c)$ in all formulas.

Problem 2.1 — SA (2 marks)

Two firms, $p = 200 - Q$, zero costs. Best response of firm 2 if $q_1 = 100$?

Solution 2.1

$BR_2(100) = \frac{200-100}{2} = \boxed{50}$

Problem 2.2 — SA (1 mark)

Same game. Nash equilibrium quantities $q_1 = q_2 =$?

Solution 2.2

$q^* = 200/3 \approx \boxed{66.67}$. (Accept range 66.5–67.0 in exam.)

Problem 2.3 — SA (1 mark)

Equilibrium profit of each firm? (Report nearest integer.)

Solution 2.3

$\pi^* = (200/3)^2 = 40000/9 \approx \boxed{4444}$

Problem 2.4 — SA (2 marks)

$p = 180 - Q$, zero costs. Find NE quantities and profit.

Solution 2.4

$q^* = 180/3 = \boxed{60}$. $\pi^* = 60^2 = \boxed{3600}$.

Problem 2.5 — SA (2 marks)

$p = 240 - Q$, $c_1 = c_2 = 30$. BR of firm 2 if $q_1 = 60$?

Solution 2.5

Effective intercept for firm 2: $a - c = 240 - 30 = 210$.
$BR_2(q_1) = \frac{210-q_1}{2} = \frac{210-60}{2} = \boxed{75}$

Problem 2.6 — SA (1 mark)

Same game ($p = 240 - Q$, $c = 30$). NE quantity?

Solution 2.6

$q^* = (240 - 30)/3 = 210/3 = \boxed{70}$

Problem 2.7 — SA (1 mark)

NE profit for each firm?

Solution 2.7

$p^* = 240 - 2(70) = 100$. $\pi^* = (100 - 30) \times 70 = 70 \times 70 = \boxed{4900}$.

Problem 2.8 — SA (2 marks)

$p = 300 - Q$, $c_1 = 20$, $c_2 = 40$. Find q_1^* and q_2^*.

Solution 2.8

$BR_1 : q_1 = (280 - q_2)/2$. $\quad BR_2 : q_2 = (260 - q_1)/2$.
Substituting: $q_1 = (280 - (260 - q_1)/2)/2 = (280 - 130 + q_1/2)/2 = (150 + q_1/2)/2$
$q_1 = 75 + q_1/4 \implies 3q_1/4 = 75 \implies q_1 = \boxed{100}$.
$q_2 = (260 - 100)/2 = \boxed{80}$.

Roommate Cleaning Game — ˜3 marks

Always the same pattern: BR → dominance → symmetric NE.

Template: Payoff $u_i = (K - t_j)t_i - t_i^2$

Best Response: $BR_i(t_j) = \dfrac{K - t_j}{2}$
Max BR (when $t_j = 0$): $BR_i(0) = K/2$. Any $t_i > K/2$ is dominated by $K/2$.
Symmetric NE: $t^* = \dfrac{K}{3}$ (write as $t_1 = t_2 = \alpha/3$, then $\alpha = K$)

Problem 2.9 — SA (1 mark)

$u_i = (10 - t_j)t_i - t_i^2$. Best response if $t_1 = 0$? $t_2 = ?$

Solution 2.9

$BR_2(0) = (10 - 0)/2 = \boxed{5}$

Problem 2.10 — MCQ (1 mark)

Any $t_i > 5$ is dominated by $t_i = 5$. True or False?

Solution 2.10

True. $BR_i(t_j) \leq 5$ for all $t_j \geq 0$, so $t_i > 5$ is never a best response.

Problem 2.11 — SA (1 mark)

Symmetric NE: $t_1 = t_2 = \alpha/3$. Value of α?

Solution 2.11

$t^* = 10/3$, so $\alpha = \boxed{10}$.

Problem 2.12 — SA (1 mark)

$u_i = (14 - t_j)t_i - t_i^2$. Symmetric NE t^*?

Solution 2.12

$t^* = 14/3 \approx \boxed{4.67}$

Problem 2.13 — SA (1 mark)

Same game. BR of player 2 if $t_1 = 4$?

Solution 2.13

$BR_2(4) = (14 - 4)/2 = \boxed{5}$

Market Entry Game — ˜3–4 marks

Pattern: find threshold n^*, count pure NE, then solve mixed NE quadratic.

Template: Payoff $= V/n$, Cost $= C$

1. Find max n^* where $V/n > C$ (profitable to enter).
2. Pure NE: exactly n^* firms enter $\Rightarrow \binom{N}{n^*}$ equilibria.
3. "All enter" is NE iff $V/N \geq C$.

4. Mixed NE: solve E[payoff from entering] $= 0$.

Problem 2.14 — SA (1 mark)

3 firms, payoff $= 150/n$, cost $= 62$. How many pure strategy NE?

Solution 2.14

$n = 1$: $150 - 62 = 88 > 0$. $n = 2$: $75 - 62 = 13 > 0$. $n = 3$: $50 - 62 = -12 < 0$.
Exactly 2 enter. $\binom{3}{2} = \boxed{3}$ pure NE.

Problem 2.15 — MCQ (1 mark)

"In one of these pure NEs, all three players enter." True or False?

Solution 2.15

FALSE. If all 3 enter, each gets $50 - 62 = -12 < 0$, so any firm deviates to "stay out."

Problem 2.16 — MCQ (2 marks)

Mixed strategy probability p of entering?

A. 1/5 B. 3/5 C. 4/5 D. 2/5

Solution 2.16

Expected payoff from entering $= 0$:

$$\binom{2}{0}(1-p)^2(88) + \binom{2}{1}p(1-p)(13) + \binom{2}{2}p^2(-12) = 0$$

$88(1-p)^2 + 26p(1-p) - 12p^2 = 0$
$88 - 176p + 88p^2 + 26p - 26p^2 - 12p^2 = 0$
$50p^2 - 150p + 88 = 0$
$p = \frac{150 \pm \sqrt{22500-17600}}{100} = \frac{150 \pm 70}{100}$
$p = 0.8$ or $p = 2.2$ (reject). **Answer: C** ($p = 4/5$)

Problem 2.17 — SA (1 mark)

4 firms, payoff $= 200/n$, cost $= 55$. How many pure NE?

Solution 2.17

$n = 1$: $145 > 0$. $n = 2$: $45 > 0$. $n = 3$: $11.7 > 0$. $n = 4$: $-5 < 0$.
Exactly 3 enter. $\binom{4}{3} = \boxed{4}$.

Problem 2.18 — SA (1 mark)

5 firms, payoff $= 250/n$, cost $= 60$. How many pure NE?

Solution 2.18

$n = 1$: 190. $n = 2$: 65. $n = 3$: 23.3. $n = 4$: $2.5 > 0$. $n = 5$: $-10 < 0$.
Exactly 4 enter. $\binom{5}{4} = \boxed{5}$.

2 Week 3: Extensive Form — Declining Industry Game

Exit Game — ˜3–5 marks

Appears in both exam papers. Learn the payoff matrix construction.

Payoff Matrix: E/T/N Exit Game

Both operating = -1/quarter each. One alone = $+2$/quarter. Exit payoff = 0 onward.

	E	T	N
E	$(0,0)$	$(0,2)$	$(0,4)$
T	$(2,0)$	$(-1,-1)$	$(-1,1)$
N	$(4,0)$	$(1,-1)$	$(-2,-2)$

Key results: No strictly dominated strategies. T is weakly dominated.
Pure NE: (E,N) and (N,E). Unique mixed NE exists.
After eliminating T: $\beta/\alpha = 2$ (where $\alpha = P(E)$, $\beta = P(N)$).

Problem 3.1 — MCQ (1 mark)

"There are no strictly dominated strategies in this game."

A. True B. False

Solution 3.1

A (True). Check: E is not dominated (E gives 0 vs T's -1 when opponent plays T, but E gives 0 vs N's 4 when opponent plays E). No strategy is beaten by another in *every* column.

Problem 3.2 — MCQ (1 mark)

The weakly dominated strategy is:

A. E B. T C. N D. None

Solution 3.2

B (T). Compare T and E for Player 1: $(2,-1,-1)$ vs $(0,0,0)$. T beats E when opponent plays E (2¿0), ties/loses elsewhere. Now compare T and N: $(2,-1,-1)$ vs $(4,1,-2)$. N weakly dominates T: $4 \geq 2$, $1 \geq -1$, $-2 \leq -1$ ... actually N doesn't dominate T in the (N,N) cell $(-2 < -1)$.
More carefully: E weakly dominates T? $(0 \leq 2, 0 \geq -1, 0 \geq -1)$ — no, $0 < 2$ in column E. But we need $u(E,\cdot) \geq u(T,\cdot)$ everywhere: $(0,0,0)$ vs $(2,-1,-1)$: fails at column E $(0 < 2)$. Actually the answer from the exam is T. The domination works by a *mixture* of E and N.

Problem 3.3 — MSQ (1 mark)

Pure strategy Nash equilibria are: (More than one correct)

A. (E,T) B. (E,N) C. (N,E) D. (T,N)

Solution 3.3

Check (E, N): P1 gets 0 (vs 0 from T, 0 from N... wait). P1 payoff at $(E, N) = 0$. If P1 deviates to T: (-1). To N: (-2). So E is BR. P2 payoff at $(E, N) = 4$. If P2 deviates to E: 0. To T: 2. So N is BR. ✓ NE.
Check (N, E): Symmetric. ✓ NE.
Answer: B and C.

Problem 3.4 — MCQ (1 mark)

"This game has a unique mixed strategy equilibrium."

A. True B. False

Solution 3.4

A (True).

Problem 3.5 — SA (2 marks)

If randomization probability of E is α and of N is β, find β/α.

Solution 3.5

After eliminating T (weakly dominated), the reduced game on $\{E, N\}$:

	E	N
E	$(0, 0)$	$(0, 4)$
N	$(4, 0)$	$(-2, -2)$

P2 indifferent: $E[u_2(E)] = E[u_2(N)]$
$0 \cdot \alpha + 0 \cdot \beta = 4\alpha + (-2)\beta$
$0 = 4\alpha - 2\beta \implies \beta/\alpha = \boxed{2}$
(So $\alpha + \beta = 1$, giving $\alpha = 1/3$, $\beta = 2/3$.)

3 Week 4: Auctions — First-Price & Order Statistics

FPSB Auction + Order Stats — ˜4–5 marks, easiest marks in the exam

Two formulas, plug and play. Never miss these.

The Only Three Formulas You Need

n bidders, valuations i.i.d. Uniform$[0, 1]$.

1. **Optimal bid in FPSB:** $b(v) = \dfrac{n-1}{n} \cdot v$
2. **Expected seller revenue:** $E[R] = \dfrac{n-1}{n+1}$
3. **Expected k-th order statistic:** $E[X_{(k)}] = \dfrac{k}{n+1}$

Problem 4.1 — MCQ (1 mark)

5 bidders, $v = 0.6$. Optimal bid?

A. 0.40 B. 0.48 C. 0.50 D. 0.55

Solution 4.1

$b = \frac{4}{5} \times 0.6 = \boxed{0.48}$. **Answer: B**

Problem 4.2 — MCQ (1 mark)

6 bidders, $v = 0.9$. Optimal bid?

A. 0.48 B. 0.65 C. 0.5 D. 0.75

Solution 4.2

$b = \frac{5}{6} \times 0.9 = \boxed{0.75}$. **Answer: D**

Problem 4.3 — MCQ (1 mark)

8 bidders, $v = 0.56$. Optimal bid?

A. 0.49 B. 0.42 C. 0.48 D. 0.50

Solution 4.3

$b = \frac{7}{8} \times 0.56 = 0.49$. **Answer: A**

Problem 4.4 — MCQ (1 mark)

5 bidders. Expected revenue of seller?

A. 4/5 B. 3/5 C. 2/3 D. 2/5

Solution 4.4

$E[R] = \frac{4}{6} = \frac{2}{3}$. **Answer: C**

Problem 4.5 — MCQ (1 mark)

6 bidders. Expected revenue?

A. 4/5 B. 5/7 C. 2/3 D. 2/5

Solution 4.5

$E[R] = \frac{5}{7}$. **Answer: B**

Problem 4.6 — MCQ (1 mark)

7 bidders. Expected revenue?

A. 3/4 B. 5/7 C. 7/8 D. 6/7

Solution 4.6

$E[R] = \frac{6}{8} = \frac{3}{4}$. **Answer: A**

Problem 4.7 — MCQ (2 marks)

5 numbers from Uniform$[0, 1]$, sorted. Expected value in 2nd position?

A. 5/6 B. 2/3 C. 1/2 D. 1/3

Solution 4.7

$E[X_{(2)}] = \frac{2}{6} = \frac{1}{3}$. **Answer: D**

Problem 4.8 — MCQ (2 marks)

6 numbers from Uniform$[0, 1]$, sorted. Expected value in 3rd position?

A. 5/6 B. 3/7 C. 1/2 D. 1/3

Solution 4.8

$E[X_{(3)}] = \frac{3}{7}$. **Answer: B**

Problem 4.9 — MCQ (2 marks)

8 numbers from Uniform$[0, 1]$, sorted. Expected 5th smallest?

A. 5/8 B. 4/9 C. 5/9 D. 1/2

Solution 4.9

$E[X_{(5)}] = \frac{5}{9}$. **Answer: C**

Problem 4.10 — SA (1 mark)

10 numbers from Uniform$[0, 1]$. Expected value of the maximum?

Solution 4.10

$E[X_{(10)}] = \frac{10}{11} \approx \boxed{0.909}$

4 Week 6: Bayes' Theorem (Defective Items)

Bayes' Theorem — 4 marks, guaranteed easy marks

Same structure every time. Drill the template.

Template

Machines A, B, C produce fractions p_A, p_B, p_C with defect rates d_A, d_B, d_C.
$P(\text{Def}) = p_A \cdot d_A + p_B \cdot d_B + p_C \cdot d_C$
$P(\text{Machine } A \mid \text{Def}) = \dfrac{p_A \cdot d_A}{P(\text{Def})}$

Problem 6.1 — SA (2 marks)

A: 50%, B: 30%, C: 20%. Defect rates: 3%, 4%, 5%. $P(\text{defective})$?

Solution 6.1

$P(D) = 0.50(0.03) + 0.30(0.04) + 0.20(0.05) = 0.015 + 0.012 + 0.010 = \boxed{0.037}$

Problem 6.2 — SA (2 marks)

Given defective, probability from machine A?

Solution 6.2

$P(A|D) = \frac{0.015}{0.037} = \boxed{0.4054} \approx 0.405$

Problem 6.3 — SA (2 marks)

A: 40%, B: 35%, C: 25%. Defect rates: 2%, 5%, 4%. $P(\text{defective})$?

Solution 6.3

$P(D) = 0.40(0.02) + 0.35(0.05) + 0.25(0.04) = 0.008 + 0.0175 + 0.010 = \boxed{0.0355}$

Problem 6.4 — SA (2 marks)

Given defective, probability from machine B?

Solution 6.4

$P(B|D) = \frac{0.0175}{0.0355} \approx \boxed{0.493}$

Problem 6.5 — SA (2 marks)

A: 60%, B: 25%, C: 15%. Defect rates: 1%, 6%, 3%. Given defective, probability from C?

Solution 6.5

$P(D) = 0.60(0.01) + 0.25(0.06) + 0.15(0.03) = 0.006 + 0.015 + 0.0045 = 0.0255$
$P(C|D) = \frac{0.0045}{0.0255} \approx \boxed{0.176}$

Exam Tip

These are always 2 + 2 = 4 marks. The computation is pure arithmetic. Write down the formula, plug in carefully, and you'll get full marks every time. Use a calculator if available.

Conceptual MCQs — 2–3 free marks

Memorize these patterns. Same questions repeat across exams.

Problem 6.6 — MCQ (1 mark)

Consider: (I) Crowd can never be wrong even if everyone is rational. (II) Aggregate behaviour of many with limited info can never produce accurate results. Which is true?

A. Only I B. Only II C. Both D. None

Solution 6.6

Both false. Cascades make crowds wrong; wisdom of crowds produces accuracy. **Answer: D**

Problem 7.1 — MCQ (1 mark)

Choose correct:

A. Stable matching = everyone satisfied

B. Stable iff an unmatched pair can beneficially deviate

C. Both A and B

D. None

Solution 7.1

A is wrong (stable $\neq$ satisfied). B describes *instability*, not stability. **Answer: D**

Problem 7.2 — MCQ (1 mark)

In $[7; 5, 1, 3, 4]$, which player is null?

A. Player 1 B. Player 2 C. Player 3 D. Player 4

Solution 7.2

Player 2 (weight 1). Adding player 2 to any coalition never changes $v(S)$. **Answer: B**

5 Week 9: Network Effects

Network Effects — 6–7 marks, multiple sub-questions

Two variants appear: $U = \theta(a + vn^e)$ and $U = \theta(a + 3\theta vn^e)$.

Variant 1: $U(\theta) = \theta(a + vn^e)$ — Standard Model

Quantity	Formula	Derivation
Indifferent consumer $\hat{\theta}$	$\dfrac{p}{a + vn^e}$	Set $U(\hat{\theta}) = p$
Max price for buyers	$a + vn^e$	Require $\hat{\theta} \leq 1$
Mass of buyers n	$1 - \hat{\theta}$	$\theta \in [\hat{\theta}, 1]$ buys
WTP $p(n^e, n)$	$(a + vn^e)(1 - n)$	Substitute $\hat{\theta}$
Fulfilled demand ($n^e\!=\!n$)	$(a + vn)(1 - n)$	Replace n^e with n
$p(n^e, n)$ is...	$\downarrow$ in n, $\uparrow$ in n^e	Partial derivatives

Problem 9.1 — MCQ (1 mark)

$\hat{\theta}$ for standard model?

A. $(p - a)/(vn^e)$ B. $p/(a + vn^e)$ C. $ap/(a + vn^e)$ D. None

Solution 9.1

B. $\hat{\theta}(a + vn^e) = p \implies \hat{\theta} = p/(a + vn^e)$.

Problem 9.2 — MCQ (1 mark)

Max p to have buyers?

A. $a + vn^e$ B. vn^e C. $a/(vn^e)$ D. $a/(1 + vn^e)$

Solution 9.2

A. $\hat{\theta} \leq 1 \implies p \leq a + vn^e$.

Problem 9.3 — MCQ (1 mark)

Mass of buyers as function of $\hat{\theta}$?

A. $\hat{\theta}$ B. $1 + \hat{\theta}$ C. $1 - \hat{\theta}$ D. $(1 + \hat{\theta})/(1 - \hat{\theta})$

Solution 9.3

C. Buyers have $\theta \geq \hat{\theta}$, so $n = 1 - \hat{\theta}$.

Problem 9.4 — MCQ (1 mark)

WTP $p(n^e, n)$?

A. $(a + vn^e)$ B. $n(a + vn^e)$ C. $(a + vn^e)(1 + n)$ D. $(a + vn^e)(1 - n)$

Solution 9.4

D.

Problem 9.5 — MSQ (1 mark)

$p(n^e, n)$ is (select all correct):

A. Increasing in n B. Decreasing in n C. Decreasing in n^e D. Increasing in n^e

Solution 9.5

B and D. $\partial p/\partial n = -(a + vn^e) < 0$; $\partial p/\partial n^e = v(1 - n) > 0$.

Problem 9.6 — MCQ (1 mark)

Fulfilled-expectations demand curve ($n^e = n$)?

A. $(a + vn)$ B. $n(a + vn)$ C. $(a + vn)(1 + n)$ D. $(a + vn)(1 - n)$

Solution 9.6

D.

Variant 2: $U(\theta) = \theta(a + 3\theta vn^e) = a\theta + 3vn^e\theta^2$ (Dec 2025)

Quantity	**Formula**
Indifferent $\hat{\theta}$	$\dfrac{p-a}{3vn^e}$ (from exam MCQ)
Max price	$a + 3vn^e$
Mass of buyers	$n = \dfrac{a - p + 3vn^e}{3vn^e}$
WTP	$a + 3vn^e(1 - n)$
Fulfilled demand ($m = 1 - n$)	$a + 3vm(1 + m)$

Problem 9.7 — MCQ (1 mark)

For $U(\theta) = a\theta + 3\theta^2 vn^e$, indifferent $\hat{\theta}$?

A. $(p - a)/(3vn^e)$ B. $p/(a + 3vn^e)$ C. $ap/(a + 3vn^e)$ D. None

Solution 9.7

A. (From exam answer key.)

Problem 9.8 — MCQ (1 mark)

Max p for buyers in variant 2?

A. $a + 3vn^e$ B. $3vn^e$ C. $a/(3vn^e)$ D. $a/(1 + 3vn^e)$

Solution 9.8

A.

Problem 9.9 — MCQ (1 mark)

WTP for the n^{th} unit in variant 2?

A. $a + 3vn^e(1 - n)$ B. $n(a + 2vn^e)$ C. $(a + 3vn^e)(1 + n)$ D. $(a + 3vn^e)(1 - n)$

Solution 9.9

A. (From exam: the correct answer for the $U = a + 3\theta vn^e$ variant.)

Problem 9.10 — MCQ (1 mark)

Fulfilled-expectations demand for variant 2 (with $m = 1 - n$)?

A. $(a + 3vm)$ B. $m(a + 3vm)$ C. $a + 3vm(1 + m)$ D. $(a + 3vm)(1 - m)$

Solution 9.10

C. (From Dec 2025 exam answer key.)

6 Week 10: Coalition Functions & Shapley-Shubik

Weighted Majority + Shapley — 8–10 marks, HIGHEST weightage

The exam always has 6–15 sub-questions on one weighted majority game. This is where you score the most marks with systematic work.

Step-by-Step Method

Given $[q; w_1, w_2, \ldots, w_n]$:

1. **Coalition function:** For each subset S, compute $\sum_{i \in S} w_i$. Set $v(S) = 1$ if $\geq q$, else 0.
2. **Symmetric players:** i, j symmetric if swapping them in every coalition preserves v.
3. **Null player:** i is null if $v(S \cup \{i\}) = v(S)$ for all S.
4. **Shapley-Shubik:** Count orderings where i is pivotal. Divide by $n!$.

Problem 10.1–10.15 — Full workout on $[61; 40, 40, 30, 10$

](**a**)*Fillinthecoalitionfunctiontable.*
(**b**)*Howmanysymmetricplayers?*
(**c**)*Whichplayerisnull?*
(**d**)*Find*1/x$_i$ for each player's Shapley-Shubik index.

Solution 10.1–10.15

(a) Coalition function:

S	Σw	$v(S)$	S	Σw	$v(S)$
$\{1\}$	40	0	$\{1,2,3\}$	110	1
$\{2\}$	40	0	$\{1,2,4\}$	90	1
$\{3\}$	30	0	$\{1,3,4\}$	80	1
$\{4\}$	10	0	$\{2,3,4\}$	80	1
$\{1,2\}$	80	1	$\{1,2,3,4\}$	120	1
$\{1,3\}$	70	1			
$\{1,4\}$	50	0			
$\{2,3\}$	70	1			
$\{2,4\}$	50	0			
$\{3,4\}$	40	0			

(b) 3 symmetric players (players 1, 2, 3). Verify: for any $S \subseteq \{4\}$, $v(S \cup \{1\}) = v(S \cup \{2\}) = v(S \cup \{3\})$.
(c) Player **4** is null. Adding 4 never changes any coalition value.
(d) With $4! = 24$ orderings, count pivots:

Ordering	Pivot?
By symmetry of players 1,2,3:	
Each of players 1,2,3 is pivotal in exactly 8 orderings.	
Player 4 is pivotal in 0 orderings.	

$\phi_1 = \phi_2 = \phi_3 = 8/24 = 1/3$. So $1/x_i = \boxed{3}$ for $i = 1, 2, 3$.

$\phi_4 = 0$, so $x_4 = \boxed{0}$.

Problem 10.16 — Full workout on $[8; 5, 4, 3, 1$

]**(a)**$Computeall$v(S).
(b)$Identifynull/symmetricplayers.$
(c)$FindShapleyvalues.$

Solution 10.16

(a):

$\{1\} = 5$	< 8	0	$\{1,2,3\} = 12$	≥ 8	1
$\{2\} = 4$		0	$\{1,2,4\} = 10$		1
$\{3\} = 3$		0	$\{1,3,4\} = 9$		1
$\{4\} = 1$		0	$\{2,3,4\} = 8$		1
$\{1,2\} = 9$		1	$\{1,2,3,4\} = 13$		1
$\{1,3\} = 8$		1			
$\{1,4\} = 6$		0			
$\{2,3\} = 7$		0			
$\{2,4\} = 5$		0			
$\{3,4\} = 4$		0			

(b): No null players (each player is pivotal somewhere). No symmetric pair (verify: $v(\{1,3\}) = 1$ but $v(\{2,3\}) = 0$, so 1 and 2 not symmetric; $v(\{1,2\}) = 1$ but $v(\{1,3\}) = 1$, need full check).
(c): Enumerate all $4! = 24$ orderings, mark pivots:
Player 1 is pivotal when preceding sum $\in [3,7]$ (adding 5 crosses 8):
Preceding sets: $\{3\} = 3$, $\{4\} = 1$ (no), $\{2\} = 4$, $\{3,4\} = 4$, $\{2,4\} = 5$, $\{2,3\} = 7$. Valid: $\{3\}, \{2\}, \{3,4\}, \{2,4\}, \{2,3\} = 5$ subsets.
Orderings: $1! \cdot 2! + 1! \cdot 2! + 2! \cdot 1! + 2! \cdot 1! + 2! \cdot 1! = 2+2+2+2+2 = 10$. $\phi_1 = 10/24 = 5/12$.
Player 2 pivotal when preceding sum $\in [4,7]$:
$\{1\} = 5$, $\{3\} = 3$ (no), $\{4\} = 1$ (no), $\{1,3\} = 8$ (no, ≥ 8!), $\{1,4\} = 6$, $\{3,4\} = 4$.
Valid: $\{1\} = 5$, $\{1,4\} = 6$, $\{3,4\} = 4 = 3$ subsets.
Orderings: $1! \cdot 2! + 2! \cdot 1! + 2! \cdot 1! = 2+2+2 = 6$. $\phi_2 = 6/24 = 1/4$.
Player 3 pivotal when preceding sum $\in [5,7]$:
$\{1\} = 5$, $\{2\} = 4$ (no), $\{1,4\} = 6$, $\{2,4\} = 5$, $\{1,2\} = 9$ (no).
Valid: $\{1\} = 5$, $\{1,4\} = 6$, $\{2,4\} = 5 = 3$ subsets.
Orderings: $2+2+2 = 6$. $\phi_3 = 6/24 = 1/4$.
Player 4 pivotal when preceding sum $\in [7,7]$ (adding 1 crosses 8):
$\{1,2\} = 9$ (no), $\{1,3\} = 8$ (no), $\{2,3\} = 7$ (yes!), $\{1,2,3\} = 12$ (no).
Valid: $\{2,3\} = 7 = 1$ subset. Orderings: $2! \cdot 1! = 2$. $\phi_4 = 2/24 = 1/12$.
Check: $5/12 + 1/4 + 1/4 + 1/12 = 5/12 + 3/12 + 3/12 + 1/12 = 12/12 = 1$. ✓

Problem 10.17 — Shapley Value (General Game)

$N = \{1,2,3\}$. $v(1) = v(2) = 5, v(3) = 0, v(1,2) = 10, v(1,3) = v(2,3) = 5, v(1,2,3) = 10$.
Find S_1, S_2, S_3 (Shapley values).

Solution 10.17

Order	mc_1	mc_2	mc_3
1-2-3	$5-0=5$	$10-5=5$	$10-10=0$
1-3-2	5	$10-5=5$	$5-5=0$
2-1-3	$10-5=5$	5	$10-10=0$
2-3-1	$10-5=5$	5	$5-5=0$
3-1-2	$5-0=5$	$10-5=5$	0
3-2-1	$10-5=5$	$5-0=5$	0
Sum	30	30	0

$S_1 = \boxed{5}$, $S_2 = \boxed{5}$, $S_3 = \boxed{0}$.

7 Week 11: Contested Garment & Cable TV

Contested Garment — 2–3 marks, pure plug-and-play

Contested Garment Formula

Estate E, claims c_1 and c_2 ($c_1 + c_2 > E$).
$\text{Concession}_1 = \max(E - c_1, 0)$ (given to Creditor 2)
$\text{Concession}_2 = \max(E - c_2, 0)$ (given to Creditor 1)
$\text{Contested} = E - \text{Conc}_1 - \text{Conc}_2$, split equally.
$x_1 = \text{Conc}_2 + \frac{\text{Contested}}{2}$, $x_2 = E - x_1$.

Problem 11.1 — SA (1+1 marks)

$E = 500$, $c_1 = 400$, $c_2 = 300$. Find x_1 and x_2.

Solution 11.1

$\text{Conc}_1 = 500 - 400 = 100$. $\text{Conc}_2 = 500 - 300 = 200$.
$\text{Contested} = 500 - 100 - 200 = 200$. Split: 100 each.
$x_1 = 200 + 100 = \boxed{300}$. $x_2 = 100 + 100 = \boxed{200}$.

Problem 11.2 — SA (1+1 marks)

$E = 500$, $c_1 = 450$, $c_2 = 250$. Find x_1 and x_2.

Solution 11.2

$\text{Conc}_1 = 500 - 450 = 50$. $\text{Conc}_2 = 500 - 250 = 250$.
$\text{Contested} = 500 - 50 - 250 = 200$. Split: 100 each.
$x_1 = 250 + 100 = \boxed{350}$. $x_2 = 50 + 100 = \boxed{150}$.

Problem 11.3 — SA (1+1 marks)

$E = 600$, $c_1 = 500$, $c_2 = 400$. Find x_1 and x_2.

Solution 11.3

$\text{Conc}_1 = 100$. $\text{Conc}_2 = 200$.
$\text{Contested} = 600 - 100 - 200 = 300$. Split: 150 each.
$x_1 = 200 + 150 = \boxed{350}$. $x_2 = 100 + 150 = \boxed{250}$.

Problem 11.4 — SA (1+1 marks)

$E = 400$, $c_1 = 350$, $c_2 = 250$. Find x_1 and x_2.

Solution 11.4

$\text{Conc}_1 = 400 - 350 = 50$. $\text{Conc}_2 = 400 - 250 = 150$.
$\text{Contested} = 400 - 50 - 150 = 200$. Split: 100.
$x_1 = 150 + 100 = \boxed{250}$. $x_2 = 50 + 100 = \boxed{150}$.

Problem 11.5 — SA (1+1 marks)

$E = 300$, $c_1 = 300$, $c_2 = 200$. Find x_1 and x_2.

Solution 11.5

$\text{Conc}_1 = 300 - 300 = 0$. $\text{Conc}_2 = 300 - 200 = 100$.
Contested $= 300 - 0 - 100 = 200$. Split: 100.
$x_1 = 100 + 100 = \boxed{200}$. $x_2 = 0 + 100 = \boxed{100}$.

Cable TV Cost Allocation — 3–6 marks

Identify edges, count how many players use each edge, divide.

Method

For each edge e in the tree from station to players:

$$\text{Player } i\text{'s cost} = \sum_{\substack{e \text{ on path} \\ \text{from station to } i}} \frac{\text{cost}(e)}{\#\text{ players whose path includes } e}$$

Problem 11.6 — SA (3 marks)

Station $\xrightarrow{18}$ Player 1 $\xrightarrow{24}$ Junction $\xrightarrow{12}$ Player 2, Junction $\xrightarrow{12}$ Player 3.
Find cost for each player.

Solution 11.6

Edge 18 (Station $\to$ P1): all 3 players use it $\to 18/3 = 6$ each.
Edge 24 (P1 $\to$ Junction): P2 and P3 use it $\to 24/2 = 12$ each.
Edge 12 (Junction $\to$ P2): only P2 $\to$ 12.
Edge 12 (Junction $\to$ P3): only P3 $\to$ 12.
$P1 = 6 = \boxed{6}$. $P2 = 6 + 12 + 12 = \boxed{30}$. $P3 = 6 + 12 + 12 = \boxed{30}$.
Check: $6 + 30 + 30 = 66 = 18 + 24 + 12 + 12 = 66$. ✓

Problem 11.7 — SA (6 marks)

6-player tree (from Dec 2025 exam):
Station $\xrightarrow{42}$ Node1 $\xrightarrow{15}$ Node2 $\xrightarrow{5}$ P4, Node2 $\xrightarrow{6}$ Node3, Node3 has P5 and P2.
Node1 $\xrightarrow{10}$ P3 $\xrightarrow{7}$ P6. P1 branches from Node1.
Find cost for all 6 players.

Solution 11.7

From the exam answers: $P1 = 7$, $P2 = 12$, $P3 = 12$, $P4 = 17$, $P5 = 18$, $P6 = 19$.
Check: $7 + 12 + 12 + 17 + 18 + 19 = 85 = 42 + 15 + 5 + 6 + 10 + 7 = 85$. ✓
The key is carefully identifying which edges each player's path uses and the number of players sharing each edge.

Problem 11.8 — SA (2 marks)

Simple tree: Station $\xrightarrow{30}$ Split $\xrightarrow{10}$ P1, Split $\xrightarrow{20}$ P2. Costs?

Solution 11.8

Edge 30: both $\rightarrow$ 15 each. Edge 10: P1 only. Edge 20: P2 only.
$P1 = 15 + 10 = \boxed{25}$. $P2 = 15 + 20 = \boxed{35}$.

8 Week 12: VCG Mechanism — Ad Auctions

VCG Ad Auctions — 5–6 marks, very systematic

Same problem appears every exam with slight number changes. Master the 3-step method.

3-Step VCG Method

Given: Slots with CTRs, advertisers with per-click values.
Step 1: Socially optimal allocation: assign highest-value advertiser to highest-CTR slot.
Total SW $= \sum \text{value}_i \times \text{CTR of assigned slot}$
Step 2: VCG price for winner of slot j (advertiser k):
(a) Remove advertiser k. Re-allocate optimally among remaining.
(b) Others' welfare WITHOUT $k = \sum_{\text{remaining}} \text{value} \times \text{CTR}$
(c) Others' welfare WITH $k = \sum_{\text{others in original}} \text{value} \times \text{their assigned CTR}$
(d) VCG price $= (b) - (c)$
Step 3: Repeat Step 2 for each slot winner.

Problem 12.1 — SA (2+2+2 marks)

Slots: a (CTR=10), b (CTR=5). Advertisers: x (3/click), y (2/click), z (1/click).
(a) Total social welfare of optimal allocation?
(b) VCG price for slot a?
(c) VCG price for slot b?

Solution 12.1

(a) Optimal: $x \to a$, $y \to b$. SW $= 3(10) + 2(5) = \boxed{40}$.
(b) Without x: $y \to a, z \to b$. Others' welfare $= 2(10) + 1(5) = 25$.
With x: others get $y \to b$. Others' welfare $= 2(5) = 10$.
Price $= 25 - 10 = \boxed{15}$.
(c) Without y: $x \to a, z \to b$. Others' welfare $= 3(10) + 1(5) = 35$.
With y: others get $x \to a$. Others' welfare $= 3(10) = 30$.
Price $= 35 - 30 = \boxed{5}$.

Problem 12.2 — SA (2+2+2 marks)

Slots: a (CTR=15), b (CTR=8). Advertisers: x (5/click), y (3/click), z (2/click).
(a) Optimal SW? (b) VCG price for a? (c) VCG price for b?

Solution 12.2

(a) $x \to a$, $y \to b$. SW $= 5(15) + 3(8) = 75 + 24 = \boxed{99}$.
(b) Without x: $y \to a$, $z \to b$. Others $= 3(15) + 2(8) = 45 + 16 = 61$.
With x: others $= 3(8) = 24$.
Price $= 61 - 24 = \boxed{37}$.
(c) Without y: $x \to a$, $z \to b$. Others $= 5(15) + 2(8) = 75 + 16 = 91$.
With y: others $= 5(15) = 75$.
Price $= 91 - 75 = \boxed{16}$.

Problem 12.3 — SA (2+2+2 marks)

Slots: a (CTR=20), b (CTR=10), c (CTR=5). Advertisers: x (4), y (3), z (2), w (1).
(a) Optimal SW? (b) VCG price for a? (c) VCG price for b?

Solution 12.3

(a) $x \to a$, $y \to b$, $z \to c$. SW $= 4(20) + 3(10) + 2(5) = 80 + 30 + 10 = \boxed{120}$.
(b) Without x: $y \to a$, $z \to b$, $w \to c$. Others $= 3(20) + 2(10) + 1(5) = 60 + 20 + 5 = 85$.
With x: others $= 3(10) + 2(5) = 30 + 10 = 40$.
Price $= 85 - 40 = \boxed{45}$.
(c) Without y: $x \to a$, $z \to b$, $w \to c$. Others $= 4(20) + 2(10) + 1(5) = 80 + 20 + 5 = 105$.
With y: others $= 4(20) + 2(5) = 80 + 10 = 90$.
Price $= 105 - 90 = \boxed{15}$.

Problem 12.4 — SA (2+2+2 marks)

Slots: a (CTR=12), b (CTR=6). Advertisers: x (4), y (3), z (1).
(a) Optimal SW? (b) VCG for a? (c) VCG for b?

Solution 12.4

(a) SW $= 4(12) + 3(6) = 48 + 18 = \boxed{66}$.
(b) Without x: $y \to a$, $z \to b$. Others $= 3(12) + 1(6) = 42$.
With x: others $= 3(6) = 18$. Price $= 42 - 18 = \boxed{24}$.
(c) Without y: $x \to a$, $z \to b$. Others $= 4(12) + 1(6) = 54$.
With y: others $= 4(12) = 48$. Price $= 54 - 48 = \boxed{6}$.

9 Quick-Reference Formula Sheet

All Key Formulas on One Page

Topic	Formula
FPSB bid	$b(v) = \frac{n-1}{n} \cdot v$
Expected revenue	$E[R] = \frac{n-1}{n+1}$
k-th order statistic	$E[X_{(k)}] = \frac{k}{n+1}$
Cournot NE ($c = 0$)	$q^* = a/3$, $\pi^* = (a/3)^2$
Cournot BR	$BR_i(q_j) = (a - c - q_j)/2$
Cournot NE ($c > 0$, symmetric)	$q^* = (a - c)/3$
Cleaning game BR	$BR_i(t_j) = (K - t_j)/2$, NE: $t^* = K/3$
Network: $\hat{\theta}$	$p/(a + vn^e)$
Network: max p	$a + vn^e$
Network: demand	$(a + vn)(1 - n)$
VCG price	Others' welfare without − Others' welfare with
Contested garment	Concessions → contested (split equally)
Cable TV cost	Edge cost / # players sharing edge
Market entry	Max n: $V/n > C$; # NE = $\binom{N}{n^*}$
Bayes	$P(A\|D) = P(D\|A)P(A)/P(D)$
Shapley-Shubik	# pivotal orderings / $n!$
Exit game NE	(E, N) and (N, E); $\beta/\alpha = 2$

Scoring Strategy — Do These First in the Exam

1. **Coalition function sub-questions** (Q2–Q7 type): ∼6 marks in 5 minutes. Pure arithmetic.
2. **Auction + Order stats**: ∼4 marks in 2 minutes. One formula each.
3. **VCG**: ∼5 marks in 5 minutes. Systematic 3-step method.

4. **Bayes**: ∼4 marks in 3 minutes. Two multiplications and one division.
5. **Contested garment**: ∼2 marks in 2 minutes.
6. **Cournot**: ∼4 marks in 3 minutes.
7. **Network effects MCQs**: ∼6 marks in 5 minutes if you know the formulas.
8. **Conceptual MCQs** (matching, crowds): ∼2 marks, 30 seconds each.

Total: ∼33 marks in 25 minutes. Then tackle market entry and exit game for the remaining 12 marks.